CHENGYU TSAI (Patrick)

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EDUCATION

NATIONAL TAIWAN UNIVERSITY **Sep. 2024 - present**

Information Management | Expected Graduation: Jun 2028

- GPA: 2025 Fall 4.20 / 4.3 | 2026 Spring 4.14 / 4.3 | Year Rank 7/49
- Relevant Coursework: Statistics (A+), Human-Computer Interaction and User Research (A+), WebApp Development (A+), Statistical Learning and Deep Learning (A+), Operating System (A).

RELEVANT PROJECT EXPERIENCE

AI LEARNING ASSISTANT PLATFORM **Jan. 2026 - June. 2026**

Project Leader | System Analysis & Integration

- Interviewed cram school staff to **identify workflow challenges and define system requirements.**
- Led **system architecture and data model design**, coordinating student, teacher, and AI components.
- Translated **business requirements into REST API** specifications and coordinated frontend, backend, and **AI integration** to deliver a functional prototype.

JOB–TALENT MATCHING CLASSIFICATION MODEL **Sep. 2025 - Dec. 2025**

Machine Learning Project | Classification & Data Analysis & Feature Engineering

- Analyzed candidate-job matching using resume data, job requirements, and structured features.
- Developed classification models using TF-IDF, feature engineering, Logistic Regression, and LightGBM.
- Improved validation **F1 score from 0.29 to 0.57** through feature engineering and threshold tuning.

FINANCIAL DATA ANALYSIS & RISK ASSESSMENT **Sep. 2025 - June. 2026**

Python-Based Financial Analysis | Statistical Modeling

- Collected and processed **financial market data** for the S&P 500, gold, U.S. Treasury bonds, Swiss franc, and Bitcoin using Python.
- Applied correlation and OLS regression to evaluate asset hedging relationships under different conditions.
- Used dummy variables and interaction terms to examine changes in asset hedging performance.

LEADERSHIP EXPERIENCE

NTU INFORMATION MANAGEMENT STUDENT ASSOCIATION **Sep. 2025 - Jun. 2026**

Finance Director

- Managed event budgets and expenses, evaluated funding needs and cost considerations, and coordinated resource allocation across student association teams.

SKILLS

Programming & Data: Python, SQL, pandas, NumPy, scikit-learn, LightGBM

Analytics & Modeling: Statistical Analysis, Feature Engineering, Machine Learning, Data Visualization

Systems & AI: REST API Design, Data Modeling, System Integration, AI API Integration

Business & Collaboration: Requirements Analysis, User Research, Project Coordination, Cross-Functional Communication